Adaptive Credit Risk Scoring with Concept Drift Detection

A Project Report Submitted by

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in partial fulfillment of the requirements for the award of the degree of

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Declaration

I hereby declare that the work presented in this Project Report titled Adaptive Credit Risk Scoring with Concept Drift Detection submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of M.Tech in Data Engineering, is a bonafide record of the research work carried out under the supervision of [Supervisor Name]. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

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Certificate

This is to certify that the Project Report titled Adaptive Credit Risk Scoring with Concept Drift Detection, submitted by Pranav Sri Vasthav Tenali Gnana (M25DE1069) to the Indian Institute of Technology Jodhpur for the award of the degree of M.Tech in Data Engineering, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

[Supervisor Name]

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I am grateful to my project supervisor, [Supervisor Name], for steady guidance across both stages of this project. The suggestions I received during the MTP-1 review, in particular the push to evaluate the static and the retrained models on exactly the same held-out data, shaped much of the experimental design reported here.

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Abstract

Credit scoring models are usually trained on one historical snapshot of borrowers and then left in service for long periods. Borrower populations, pricing policy and the economy keep moving, so the joint distribution of applicant features and repayment outcomes that the model learned slowly stops describing the loans it is asked to score. This project studies that problem, known as concept drift, on 919,696 resolved Lending Club loans issued between 2015 and 2018, and builds an end-to-end pipeline that detects drift, decides when to retrain, retrains on recent data and checks that the model's explanations remain trustworthy afterwards.

In the first stage (MTP-1) the data pipeline, three baseline classifiers and univariate drift monitoring were built. An XGBoost model trained once on 2015 loans loses 9.8% of its AUC-ROC by 2018 (0.7422 to 0.6695) while the default rate climbs from 20.5% to 27.2%. In the second stage (MTP-2) the monitor was extended with Jensen-Shannon distance and a classifier two-sample test, which exposes strong multivariate drift (domain classifier AUC of 0.977 to 0.999) even in a year where the Population Stability Index flags no single feature. Drift-triggered retraining on a three-year sliding window, scored on the same held-out test split as the static model, raises AUC-ROC by 0.018, 0.031 and 0.053 in 2016, 2017 and 2018, and PR-AUC by up to 0.074. A comparison of retraining strategies shows that recency matters more than volume: training only on the latest year gives the best 2018 AUC (0.7297), while keeping all history gives the weakest adaptive result (0.7154). SHAP-based monitoring shows that global feature importance stays stable (cosine similarity above 0.97) although the membership of the top ten features changes as interest rate grows in influence. The work is packaged as a reproducible, MLflow-tracked pipeline with an interactive Streamlit dashboard for live drift checks.

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Adaptive Credit Risk Scoring with Concept Drift Detection

1 Introduction and background

1.1 Credit scoring in consumer lending

A credit score is an estimate of how likely a borrower is to fail to repay a loan. Lenders use it to approve or reject applications, to set interest rates, and to decide how much capital to hold against a portfolio. For decades the standard tool was a logistic regression scorecard built on a small set of bureau variables. Over the last ten years gradient-boosted decision trees such as XGBoost and LightGBM have become common because they capture non-linear interactions between income, leverage, credit history and loan terms, and in public benchmarks they usually rank borrowers better than linear scorecards [8].

Almost every credit model, whatever its algorithm, is built in the same way. A set of loans with known outcomes is collected, a classifier is fitted, it is validated on a held-out sample drawn from the same period, and it is then deployed. The quiet assumption behind this workflow is that future applicants will look like past ones and will behave like them. In a live lending business that assumption rarely holds for long.

1.2 Concept drift in credit portfolios

Concept drift is the name given to any change over time in the joint distribution P(X, y) of inputs and labels [1], [2]. It helps to separate three forms. *Covariate drift* is a change in P(X): the mix of applicants changes, for example when a platform starts marketing to borrowers with higher revolving balances. *Prior or label drift* is a change in P(y): the overall default rate rises or falls with the economic cycle. *Real concept drift* is a change in P(y | X): the same profile now carries a different risk, for instance because interest rates rose and repayments became harder for a given income. The three forms usually arrive together in credit data, and all of them erode a model that was fitted once [15].

The danger is that this erosion is silent. A model keeps returning confident probabilities even when its ranking of borrowers has become worse, and true outcomes arrive only after a loan has matured or defaulted, which for consumer instalment loans can take one to five years. By the time the loss shows up in realised defaults, many loans have already been approved on a stale model. A practical credit system therefore needs to watch the inputs it receives, raise an alarm when they change, retrain when it is justified, and still explain its decisions to reviewers and regulators after each retrain.

1.3 Motivation for this project

Much of the published work on drift uses synthetic streams such as SEA or rotating hyperplanes, where the drift point is known by construction. Real credit data is messier: drift is gradual, several features move at once, labels are delayed and classes are imbalanced. The Lending Club public release

is a useful test bed because it covers several years of loans issued by one platform with consistent fields. In the 2015 to 2018 slice used here the default rate rises by almost seven percentage points and interest rates and revolving utilisation shift noticeably, so there is real, measurable drift to detect and to recover from.

The project also has an engineering motivation. Drift detection, retraining and explainability are often studied separately. A lender needs them joined into one loop with an honest evaluation protocol: a retrained model is only better if it is better on the same unseen loans that the old model is judged on. Building that loop, and measuring what each part actually contributes, is the aim of this work.

1.4 Scope of MTP-1 and MTP-2

The work was carried out in two stages. MTP-1 covered the literature study, the formulation of the problem, acquisition and cleaning of the Lending Club data, construction of yearly windows with a shared encoding, three baseline classifiers, and univariate drift monitoring with the Population Stability Index and the Kolmogorov-Smirnov test. Its main output was a measured estimate of how quickly a static model degrades. MTP-2 built on this base: it added multivariate drift detection, a leakage-free adaptive retraining protocol, a comparison of retraining strategies, threshold-aware evaluation, SHAP stability monitoring, backtesting and an interactive dashboard. This report presents both stages together.

1.5 Organisation of the report

Chapter 2 reviews prior work on credit scoring, drift detection, adaptive learning and explainability, and identifies the gaps this project addresses. Chapter 3 states the problem formally and lists the objectives. Chapter 4 describes the data, the system architecture and each method in detail. Chapter 5 reports the experimental findings, including the retraining-strategy comparison and the explainability analysis, together with implementation issues found and fixed during testing. Chapter 6 summarises the work, its limitations and the plan for future work. The appendix lists the configuration and the steps needed to reproduce every result.

2 Literature survey

2.1 Statistical and machine learning credit models

Thomas [16] surveys the classical approach to application and behavioural scoring, in which logistic regression on binned variables produces an additive scorecard that is easy to audit. Lessmann et al. [8] later benchmarked more than forty classifiers on eight retail credit datasets and found that ensembles, particularly boosted and bagged trees, consistently beat the logistic baseline, although the gains were modest and depended on the dataset. XGBoost [3] and LightGBM [4] made gradient boosting fast enough for portfolios with millions of rows and are now the usual choice for tabular risk models. None of these studies, however, examines what happens to model quality after deployment; they evaluate on random splits drawn from the same period as the training data.

2.2 Concept drift theory and detection

Gama et al. [1] and Lu et al. [2] give broad surveys of learning under drift. They separate drift *detection*, which decides whether the data has changed, from drift *adaptation*, which decides how the model should respond. Moreno-Torres et al. [15] provide a common vocabulary for covariate shift, prior shift and concept shift. Error-based detectors such as DDM [14] and ADWIN [13] monitor the running error rate of a classifier and signal when it rises significantly. They work well on streams where labels arrive immediately, but credit labels arrive months or years late, so error-based detection reacts too slowly for lending.

Distribution-based detectors avoid the label delay by comparing the inputs directly. In the credit industry the Population Stability Index is the traditional choice; Yurdakul [9] studies its statistical behaviour and the familiar 0.1 and 0.2 rule-of-thumb thresholds. The two-sample Kolmogorov-Smirnov test and the Jensen-Shannon divergence [12] are other common univariate measures. Rabanser et al. [11] compared many shift detectors and found that univariate tests applied feature by feature can miss shifts that are spread across several features, while a learned domain classifier is often more sensitive. Lopez-Paz and Oquab [10] formalised this idea as the classifier two-sample test (C2ST): if a classifier can tell reference rows from current rows better than chance, the two distributions differ.

2.3 Adaptive retraining

The simplest adaptation strategies retrain periodically on a sliding window of recent data or on all data seen so far. Sliding windows forget old concepts but may discard useful history; expanding windows keep everything but are slow to react. Drift-triggered retraining ties the retraining decision to a detector, which saves computation when the data is stable but makes the system only as good as its detector [1]. Sculley et al. [21] point out that such feedback loops are a major source of hidden technical debt in production machine learning, which argues for keeping the retraining logic explicit, versioned and tested.

2 Literature survey

2.4 Explainability of credit decisions

Lenders must explain adverse decisions to applicants and regulators. SHAP [5] attributes each prediction to its input features using Shapley values, and TreeSHAP [6] computes exact values for tree ensembles in polynomial time. Most work computes explanations for a single fitted model. When a model is retrained, its explanations may change as well, and a reviewer needs to know whether the reasoning behind decisions has shifted. Measuring the stability of explanations across time is much less studied than measuring accuracy. The closest prior work to this project, Shivogo [18], combines drift-aware credit scoring with fairness and SHAP explanations and motivated several choices made here.

Approach	Core idea	Limitation for credit data	Addressed here by
Static scorecard / GBM [3], [4], [8]	Train once on a snapshot	Accuracy decays unnoticed as data drifts	Yearly monitoring and backtest
Error-based detectors (DDM, ADWIN) [13], [14]	Watch running error rate	Need immediate labels; credit labels are delayed	Label-free input monitoring
PSI and KS tests [9]	Per-feature distribution shift	Blind to multivariate drift	Added JS distance and C2ST
Domain classifier / C2ST [10], [11]	Learn to separate two samples	Rarely applied to credit portfolios	Used as a retraining trigger
Sliding-window retraining [1]	Retrain on recent data	Window size chosen by assumption	Benchmarked against two baselines
SHAP explanations [5], [6]	Attribute each prediction	Stability over retrains not tracked	Cosine, Kendall tau, Jaccard@10

Table 2.1: Comparison of prior approaches and the gaps addressed in this project.

Two gaps stand out from this review. First, most drift studies on credit data rely on univariate statistics, and the literature on shift detection suggests that these can badly understate the true amount of change. Second, adaptive retraining is usually shown to help without a fair comparison: the retrained model is often evaluated on a different test set from the static model, and simpler retraining policies are rarely tested as baselines. This project targets both gaps and adds explicit tracking of explanation stability across the retraining cycle.

3 Problem definition and Objective

3.1 Problem statement

Let $D_t = \{(x_i, y_i)\}$ denote the loans issued in period t, where x_i is the vector of application features and $y_i = 1$ if the loan was charged off or defaulted. A scoring model f_o is trained on a reference period t_o and used to produce default probabilities for later periods. Concept drift exists when $P_t(x, y)$ differs from $P_{to}(x, y)$, and the AUC-ROC of f_o on D_t then falls as t moves away.

The problem addressed is to design a credit scoring pipeline that, for each new period t, (a) decides from the incoming feature and label distributions whether meaningful drift has occurred, (b) retrains the model when it has, using only data that would have been available at that time, (c) proves any benefit by scoring the static and the retrained model on the same unseen loans, and (d) reports whether the model's explanations have remained stable, so that a human reviewer can trust the retrained model.

3.2 Research questions

- **RQ1.** How much does a credit model trained once degrade over a multi-year horizon on real lending data?
- **RQ2.** Do univariate drift statistics capture the drift, or is a multivariate test needed?
- **RQ3.** Does drift-triggered retraining recover the lost performance under a leakage-free evaluation, and which retraining window works best?
- **RQ4.** Do the model's explanations stay stable as the data drifts, and can explanation monitoring act as an additional, interpretable warning signal?

3.3 Objectives

The specific objectives of the project are:

- To build a clean, reproducible data pipeline that turns the raw Lending Club release into yearly windows with an identical feature schema.
- To train and compare baseline classifiers and to quantify the degradation of a static model across 2015 to 2018.
- To implement and compare four drift detectors (PSI, KS on the label, Jensen-Shannon distance and a classifier two-sample test) and to derive a retraining trigger from them.
- To implement drift-triggered adaptive retraining and compare sliding, most-recent and expanding windows against the static model on identical held-out data.
- To report threshold-free and threshold-tuned metrics, monitor SHAP explanation stability across retrains, and deliver the loop as an MLflow-tracked pipeline with a dashboard.

3.4 Assumptions and scope

Only loans with a resolved outcome are used. Time is split into calendar years of issue, with 2015 as the reference year. Only application-time features are used, since post-origination fields such as recoveries would leak the outcome. Fairness analysis is left to future work.

4 Methodology

This chapter describes the data, the overall architecture of the system and each method used. Sections 4.2 to 4.4 were completed in MTP-1; sections 4.5 to 4.9 were developed in MTP-2. Table 4.3 at the end of the chapter summarises the division of work between the two stages.

4.1 System architecture

The system is organised as a chain of independent stages, shown in Figure 4.1. Each stage reads the artefacts written by the previous one and writes its own, so any stage can be re-run or replaced without touching the others. Configuration such as feature lists, hyperparameters, drift thresholds and window size is kept in a single YAML file. The pipeline is run through four driver scripts (one per phase after preprocessing), every training run is logged to MLflow [17], and a DVC pipeline file records the dependencies between stages.

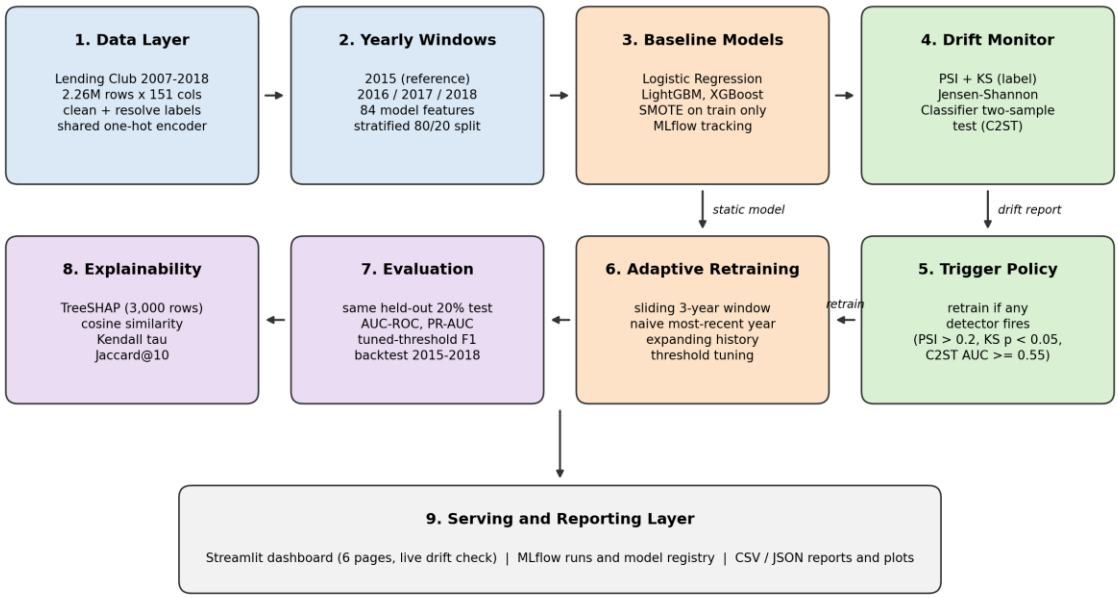


Figure 4.1: Architecture of the adaptive credit risk scoring pipeline.

4.2 Dataset and preprocessing

The source data is the public Lending Club loan file covering loans issued from 2007 to 2018 [19]. The raw file has 2,260,701 rows and 151 columns. The public release stops at the end of 2018, so no loans from 2019 onward are available. The target is derived from the loan status: Fully Paid is labelled 0, while Charged Off, Default and the two Late categories are labelled 1. Loans still Current or In Grace Period are dropped because their outcome is unknown.

Twenty application-time features were retained: twelve numerical (loan amount, interest rate, instalment, annual income, debt-to-income ratio, open accounts, public records, revolving balance, revolving utilisation, total accounts, mortgage accounts and public-record bankruptcies) and eight categorical (term, grade, sub-grade, employment length, home ownership, verification status, purpose

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and application type). Percent strings such as "13.99%" were converted to numbers, employment length was mapped to years, rows missing more than 30% of fields were dropped, and remaining gaps were filled with the median for numerical columns and an "Unknown" level for categorical ones. After restricting to issue years 2015 to 2018 the cleaned table has 919,696 rows.

The data was then split into one window per issue year. A subtle problem appeared at this step: if each window is one-hot encoded independently, a category that is absent from one year changes which level is dropped as the baseline, and the columns of different windows no longer mean the same thing. To avoid this, the category schema is fitted once on the full cleaned table and applied to every window, which gives all windows exactly the same 84 model features. Table 4.1 and Figure 4.2 summarise the windows.

Issue year	Resolved loans	Defaults	Default rate	Role
2015	377,184	77,442	20.53%	Reference / static training
2016	298,553	73,700	24.69%	Monitored period
2017	178,817	48,665	27.21%	Monitored period
2018	65,142	17,698	27.17%	Monitored period
Total	919,696	217,505	23.65%	

Table 4.1: Yearly windows after cleaning and encoding.

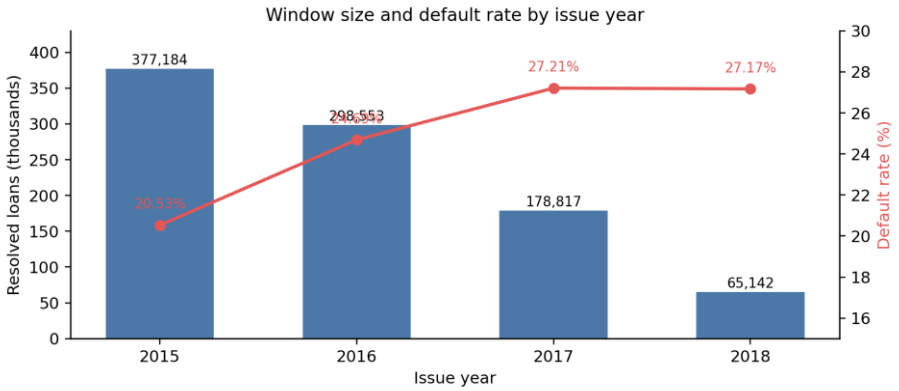


Figure 4.2: Number of resolved loans and default rate in each yearly window.

Two properties of the windows matter for the rest of the study. The default rate rises steadily, which is a built-in prior shift. The 2017 and 2018 windows are also smaller, partly because fewer loans from those years had reached a final outcome when the data was published; this label-maturity effect is discussed as a limitation in Chapter 6.

4.3 Baseline models and class imbalance

Three classifiers were trained on the 2015 window: logistic regression, LightGBM and XGBoost. Each window is split 80/20 with stratification on the target and a fixed random seed of 42. Because only about one loan in five defaults, SMOTE [7] is applied to the training portion only, which raises the 2015 training set from 301,747 to 479,586 rows with balanced classes. The test portion is never resampled,

4 Methodology

so all reported metrics reflect the real class balance. The main hyperparameters are listed in Table 4.2; they were fixed in advance and kept identical for the static and the retrained models so that any difference comes from the training data alone.

Model	Key hyperparameters
XGBoost	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column sample 0.8
LightGBM	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column sample 0.8
Logistic regression	L2 penalty, C = 1.0, max 1,000 iterations
C2ST classifier	Histogram GBM, 150 iterations, depth 4, learning rate 0.1, 5-fold CV

Table 4.2: Model hyperparameters (identical for static and adaptive models).

4.4 Univariate drift detection: PSI and KS

For each numerical feature, the Population Stability Index compares the share of the current window that falls into each of B = 10 bins with the share of the reference window. Bin edges are the deciles of the 2015 distribution, and a small constant replaces empty bins to avoid the logarithm of zero:

$$PSI = \Sigma_{i=1..B} (a_i - e_i) \cdot \ln(a_i / e_i) \tag{4.1}$$

where e_i and a_i are the expected (reference) and actual (current) proportions in bin i. The usual reading is that a PSI below 0.1 means no meaningful change, 0.1 to 0.2 a moderate change, and above 0.2 a significant shift; this project treats 0.2 as critical. Label drift is tested with the two-sample Kolmogorov-Smirnov statistic on the default indicator,

$$D_{KS} = \sup_x | F_{ref}(x) - F_{cur}(x) | \tag{4.2}$$

with drift declared when the p-value falls below $\alpha = 0.05$. These two tests formed the monitor built in MTP-1.

4.5 Multivariate drift detection: JS distance and C2ST

In MTP-2 two further detectors were added. The Jensen-Shannon distance [12] is a bounded, symmetric alternative to PSI. It is computed on the same reference deciles using base-2 logarithms, so it lies between 0 and 1:

$$JS(P, Q) = \sqrt{[\ \tfrac{1}{2}\, KL(P \,\|\, M) + \tfrac{1}{2}\, KL(Q \,\|\, M)\ ]}, \quad M = \tfrac{1}{2}(P + Q) \tag{4.3}$$

The classifier two-sample test [10], [11] addresses the main weakness of all per-feature statistics: they look at one column at a time. Up to 20,000 rows are drawn from each of the reference and current windows, labelled 0 and 1 respectively, and a histogram gradient boosting classifier from scikit-learn [20] is trained to tell them apart using all 84 features. Its five-fold cross-validated AUC is the drift score. An AUC near 0.5 means the two samples are indistinguishable; values approaching 1.0 mean the joint distribution has changed enough to identify which year a loan came from. Drift is flagged when the AUC reaches 0.55. To see which features drive the separation, the same test is also run on each feature alone.

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The retraining trigger combines the detectors. A retrain is requested for period t when any detector fires:

$$retrain_t = [\max PSI > 0.2] \vee [p_{KS} < 0.05] \vee [AUC_{C2ST} \geq 0.55] \quad (4.4)$$

The report also records what each detector would have decided on its own, which makes it possible to compare trigger policies after the fact.

4.6 Leakage-free adaptive retraining

An early version of the retraining code evaluated the adaptive model on a random 20% split of the concatenated training years while the static model was scored on the entire target year. The two numbers were therefore computed on different loans and could not be compared. The protocol used in this report fixes that. For each target year t the 20% stratified test split is set aside first. The adaptive model is trained on the earlier years in its window plus only the training portion of year t, and both the static 2015 model and the adaptive model are then scored on the same held-out test loans. No test loan is ever seen during training.

Three windowing strategies are compared. The *sliding* strategy, which is the deployed configuration, uses at most three years: [2015, 2016] for 2016, [2015, 2016, 2017] for 2017 and [2016, 2017, 2018] for 2018. The *naive recent* strategy uses only the training portion of the target year. The *expanding* strategy uses all history from 2015 up to the target year. All three use the same XGBoost settings and SMOTE on their training data.

4.7 Evaluation metrics and threshold tuning

AUC-ROC is the headline metric because it measures ranking quality independently of any cut-off, which is what a lender uses to order applicants. Because defaults are the minority class, PR-AUC (average precision) is also reported; it is more sensitive to performance on the positive class [22]. F1 at a fixed threshold of 0.5 turned out to be misleading, because models trained on SMOTE-balanced data are calibrated to a 50% default rate while the real rate is 20 to 27%. The decision threshold is therefore tuned on a validation slice holding 15% of each target year's training portion, at its real prevalence, and applied unchanged to the test set:

$$\tau^* = \operatorname{argmax}_\tau F1(y_{val}, 1[\hat{p}_{val} \geq \tau]) \quad (4.5)$$

Both the fixed-threshold and the tuned-threshold F1 are reported so that the effect of the cut-off is visible rather than hidden.

4.8 Explainability stability

TreeSHAP [6] values are computed for a random sample of 3,000 loans from each window, since exact values on the full 377,000-row reference window are too slow. The mean absolute SHAP value of each feature gives a global importance vector φ_t. Three measures compare φ_t with the reference vector φ_{ref}: cosine similarity (overall shape of the importance profile), Kendall's rank correlation τ (agreement

in feature ordering) and the Jaccard index of the top ten features (whether the same features dominate):

$$\cos(\varphi_{ref}, \varphi_t) = \varphi_{ref} \cdot \varphi_t / (\|\varphi_{ref}\| \|\varphi_t\|), \quad J@10 = |T_{ref} \cap T_t| / |T_{ref} \cup T_t| \tag{4.6}$$

Explanations are treated as stable when cosine similarity is at least 0.95 and Kendall's τ is at least 0.8. The Jaccard index is reported as an early-warning signal for changes at the top of the ranking.

4.9 MLOps tooling and dashboard

Every training run logs its parameters, metrics and model file to MLflow, with a fallback to plain artefact logging if the model registry is unavailable. Results are written as CSV and JSON reports and PNG plots, and a consolidated text summary is regenerated at the end of each run. A six-page Streamlit dashboard reads these artefacts and presents an overview, drift detection results, the static versus adaptive comparison, the retraining strategies and the explainability analysis. Its sixth page runs the full drift monitor and both models live on any built-in window or on an uploaded CSV file, reindexing the upload to the 84 reference columns.

Stage	Work carried out
MTP-1	Literature study and problem formulation; data acquisition and cleaning; yearly windows with a shared encoder; logistic regression, LightGBM and XGBoost baselines with SMOTE and MLflow; PSI and KS drift monitoring; measurement of static-model degradation.
MTP-2	Jensen-Shannon distance and classifier two-sample test; combined retraining trigger; leakage-free adaptive retraining; sliding, naive-recent and expanding strategy benchmark; PR-AUC and tuned-threshold F1; SHAP stability monitoring; backtesting; Streamlit dashboard and final report.

Table 4.3: Division of work between MTP-1 and MTP-2.

5 Theoretical/Numerical/Experimental findings

5.1 Baseline model performance

Table 5.1 reports the three baselines on the held-out 20% test split of the 2015 window. The two boosted models tie on AUC-ROC at 0.7376 and are clearly ahead of logistic regression. XGBoost was chosen as the model for all later experiments. The low F1 values at the default 0.5 threshold are an artefact of training on balanced data and are revisited in Section 5.5.

Model	AUC-ROC	F1 @ 0.5
Logistic regression	0.7299	0.2341
LightGBM	0.7376	0.2578
XGBoost (selected)	0.7376	0.2574

Table 5.1: Baseline models on the 2015 held-out test split.

5.2 Degradation of the static model (RQ1)

The XGBoost model trained on 2015 was then scored on every later window without retraining. Table 5.2 shows the result on each full window. AUC-ROC falls from 0.7422 to 0.6948 in 2016, 0.6742 in 2017 and 0.6695 in 2018, a relative loss of 9.8% over three years. The loss is largest in the first year and then continues more slowly, which matches the pattern of drift measured in the next section. F1 at 0.5 actually rises over the same period, but only because the default rate rises; this is the first sign that fixed-threshold F1 is not a reliable guide here.

Year	Loans	Default rate	AUC-ROC	F1 @ 0.5	AUC drop	Drop %
2015	377,184	20.53%	0.7422	0.2635	0.0000	0.00%
2016	298,553	24.69%	0.6948	0.3508	0.0474	6.39%
2017	178,817	27.21%	0.6742	0.4016	0.0680	9.16%
2018	65,142	27.17%	0.6695	0.4168	0.0727	9.80%

Table 5.2: Static 2015 XGBoost model scored on each full yearly window.

5.3 Drift detection results (RQ2)

Table 5.3 lists the PSI of each numerical feature against 2015. In 2016 no feature crosses the critical threshold, although interest rate is already close to the warning level at 0.095. In 2017 interest rate becomes critical (0.202). By 2018 two features are critical: revolving utilisation at 0.296 and interest rate at 0.242, with revolving balance approaching the warning band. Figure 5.1 shows the same trend for the most affected features. The KS test detects label drift in every year (statistic 0.042, 0.067 and 0.066, $p < 0.001$), reflecting the rise in the default rate.

5 Theoretical/Numerical/Experimental findings

Feature	2016	2017	2018
revol_util	0.0223	0.0876	0.2962 *
int_rate	0.0946	0.2018 *	0.2419 *
revol_bal	0.0070	0.0255	0.0963
installment	0.0098	0.0315	0.0492
dti	0.0011	0.0067	0.0471
loan_amnt	0.0062	0.0285	0.0423
total_acc	0.0023	0.0138	0.0299
open_acc	0.0003	0.0044	0.0261
mort_acc	0.0044	0.0120	0.0110
annual_inc	0.0037	0.0043	0.0076
Mean PSI	0.0129	0.0352	0.0715

Table 5.3: Population Stability Index per numerical feature (reference 2015).

Values marked * exceed the critical threshold of 0.2.

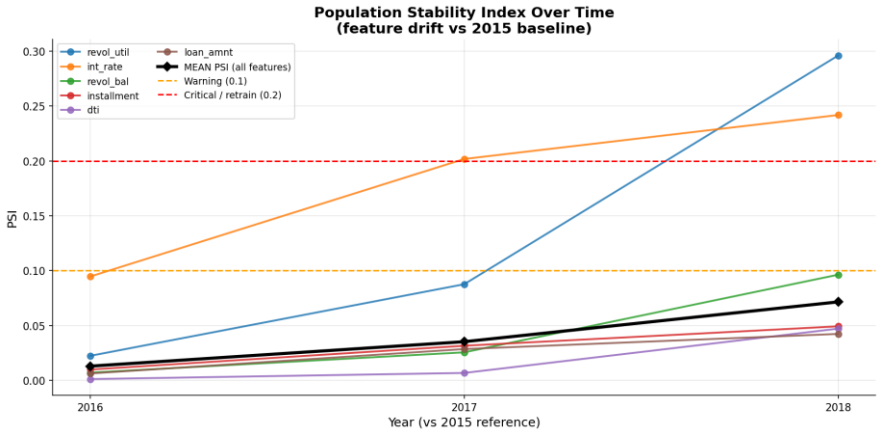


Figure 5.1: PSI of the most drifted features relative to 2015, with warning and critical thresholds.

Table 5.4 compares all four detectors. The mean Jensen-Shannon distance rises from 0.036 to 0.093 and ranks the features in almost the same order as PSI, with interest rate and revolving utilisation on top. The classifier two-sample test gives a very different picture of the size of the drift. Its AUC is already 0.977 in 2016 and reaches 0.999 in 2018, meaning a classifier can tell a 2018 loan from a 2015 loan almost perfectly. Yet the best single feature separates the two years with an AUC of only 0.54 in 2016 and 0.65 in 2018 (revolving utilisation in both cases). The drift is therefore spread across many features and their interactions, and per-feature statistics see only a small part of it.

Year	Mean PSI	Critical PSI	KS label drift	Mean JS	C2ST AUC	Retrain
2016	0.0129	0	Yes	0.0364	0.9769	Yes
2017	0.0352	1	Yes	0.0634	0.9977	Yes
2018	0.0715	2	Yes	0.0932	0.9989	Yes

Table 5.4: Comparison of drift detectors and the retraining decisions they imply.

5 Theoretical/Numerical/Experimental findings

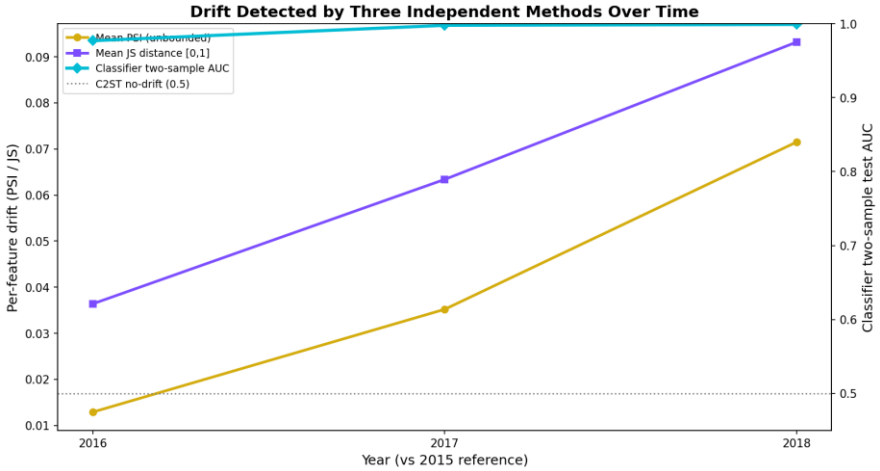


Figure 5.2: Mean PSI, mean JS distance and classifier two-sample AUC over time.

The trigger comparison has a practical consequence. A PSI-only policy would have skipped retraining in 2016, even though the static model had already lost 6.4% of its AUC that year. The KS and C2ST policies both fire in all three years. The answer to RQ2 is therefore that univariate statistics on this data under-report the drift, and a multivariate test is needed to trigger retraining at the right time.

5.4 Static versus adaptive model (RQ3)

Table 5.5 compares the static 2015 model with the drift-triggered sliding-window model. Both are scored on the identical held-out test split of each target year. The adaptive model is better in every year and on both metrics, and its advantage grows as drift accumulates: the AUC gain is 0.018 in 2016, 0.031 in 2017 and 0.053 in 2018. In 2018 the adaptive model reaches 0.7222, close to the 0.7376 that the original model achieved on its own year, so most of the lost discrimination is recovered. The PR-AUC gain is larger in relative terms, from 0.402 to 0.476 in 2018, an improvement of 18%.

Year	Static AUC	Adaptive AUC	AUC gain	Static PR-AUC	Adaptive PR-AUC	PR-AUC gain
2016	0.6941	0.7123	+0.0182	0.4002	0.4405	+0.0403
2017	0.6769	0.7081	+0.0312	0.4106	0.4551	+0.0445
2018	0.6691	0.7222	+0.0531	0.4015	0.4756	+0.0741

Table 5.5: Static and adaptive XGBoost on the same held-out test split.

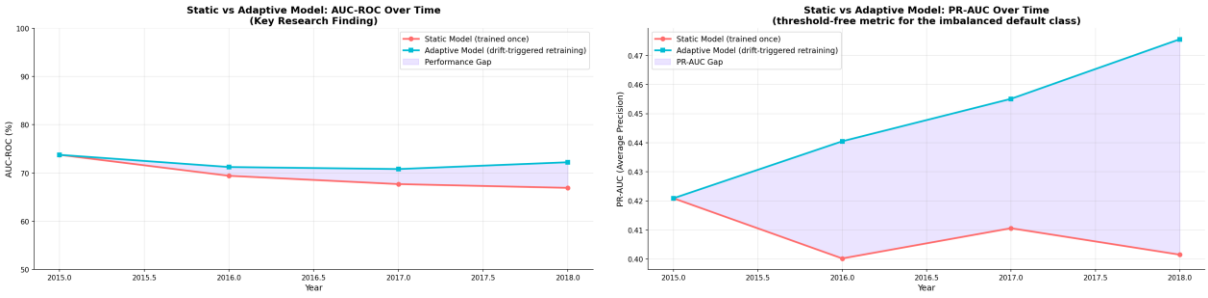


Figure 5.3: Backtest of AUC-ROC (left) and PR-AUC (right), static vs adaptive.

5 Theoretical/Numerical/Experimental findings

5.5 Effect of the decision threshold

At the fixed 0.5 threshold the static model appears to win on F1 in every year (Table 5.6). This is misleading. The adaptive model is trained on balanced data from years with more defaults, so its scores sit in a different range, and a cut-off of 0.5 labels too few loans as risky. When each model's threshold is tuned on a validation slice at the real default rate, the ordering reverses and the adaptive model has the higher F1 in every year, reaching 0.517 against 0.476 in 2018. Figure 5.4 shows both versions. Reporting only F1 at 0.5 would have led to the wrong conclusion about which model is better.

Year	Static F1 @ 0.5	Adaptive F1 @ 0.5	Static F1 tuned	Adaptive F1 tuned
2015	0.2574	0.2574	0.4356	0.4356
2016	0.3473	0.2125	0.4586	0.4840
2017	0.4026	0.2183	0.4810	0.5066
2018	0.4148	0.2477	0.4761	0.5166

Table 5.6: F1 score at the fixed 0.5 threshold and at the validation-tuned threshold.

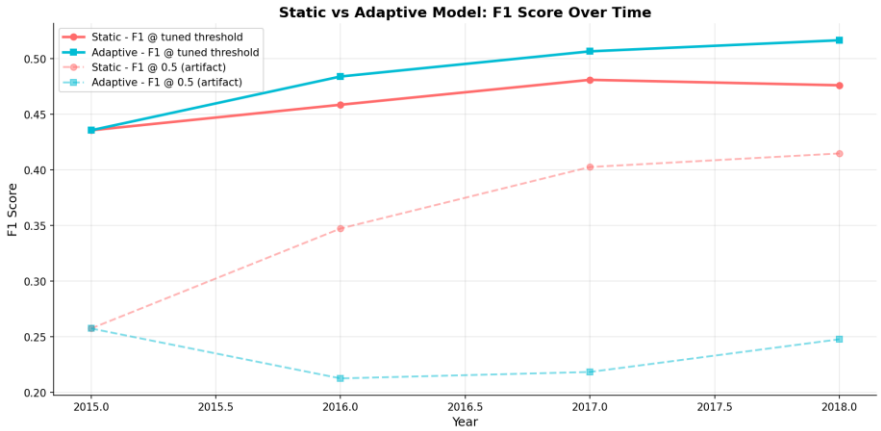


Figure 5.4: F1 of both models at the tuned threshold (solid) and at 0.5 (dashed).

5.6 Comparison of retraining strategies

Table 5.7 answers whether the sliding window is the right design by comparing it with two simpler policies. All three retraining strategies beat the static model by a wide margin in every year. Among them, the naive policy that trains only on the most recent year is the best in every year, reaching 0.7297 AUC and 0.4875 PR-AUC in 2018. The expanding window, which keeps all history, is the weakest by 2018 (0.7154). In 2016 and 2017 the sliding and expanding windows contain the same years, which is why their scores match.

Year	Static	Naive recent	Sliding (3 yr)	Expanding
2016	0.6941 (0.400)	0.7135 (0.442)	0.7123 (0.441)	0.7123 (0.441)
2017	0.6769 (0.411)	0.7128 (0.462)	0.7081 (0.455)	0.7081 (0.455)
2018	0.6691 (0.402)	0.7297 (0.488)	0.7222 (0.476)	0.7154 (0.464)

Table 5.7: AUC-ROC (PR-AUC in brackets) of each retraining strategy on the same test split.

5 Theoretical/Numerical/Experimental findings

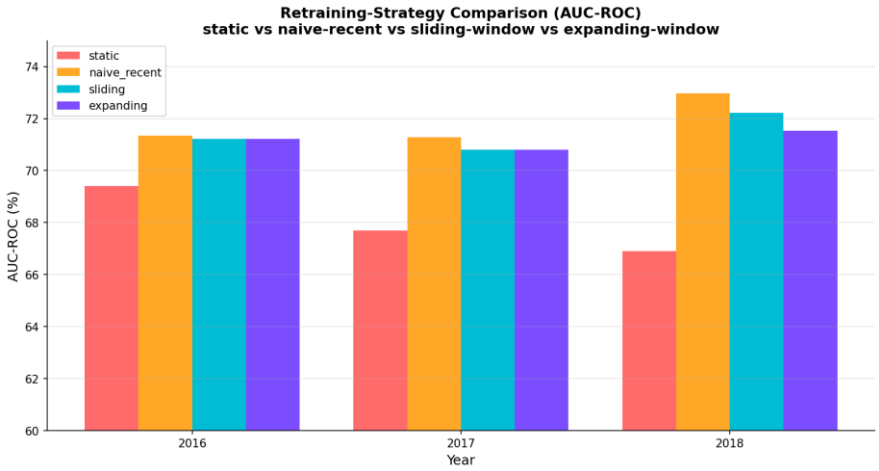


Figure 5.5: AUC-ROC of the static model and the three retraining strategies.

The finding is that recency matters more than volume. Under drift, older loans describe a borrower population and a pricing regime that no longer apply, and adding them dilutes the signal from recent loans. The sliding window is still a reasonable production default: it stays close to the best strategy, bounds the training cost and memory, and does not require knowing the ideal look-back in advance. On this data, however, a shorter window would have been better, and the window length should be treated as a tunable parameter. The answer to RQ3 is that drift-triggered retraining clearly recovers performance, with the most recent data doing most of the work.

5.7 Explainability stability (RQ4)

Table 5.8 compares the SHAP importance vector of the static model on each later year with its vector on 2015. Cosine similarity stays above 0.97 and Kendall's τ above 0.87 in every year, so by the chosen thresholds the overall explanation profile is stable. The top-ten Jaccard index tells a more detailed story: it stays at 1.0 in 2016 but falls to 0.82 in 2017 and 2018, meaning that one feature leaves the top ten and another enters. The newcomer is interest rate, which climbs into the top ten in the same years in which its PSI crosses the critical threshold.

Year	Cosine similarity	Kendall's τ	Jaccard@10	Stable
2016	0.9931	0.9043	1.0000	Yes
2017	0.9806	0.8784	0.8182	Yes
2018	0.9784	0.8827	0.8182	Yes

Table 5.8: SHAP stability of the static model relative to 2015.

5 Theoretical/Numerical/Experimental findings

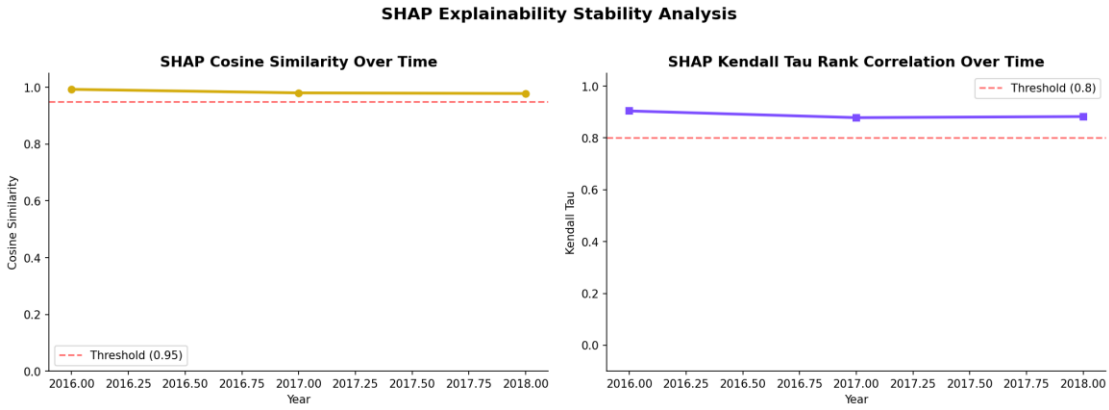


Figure 5.6: Cosine similarity and Kendall rank correlation of SHAP importance over time.

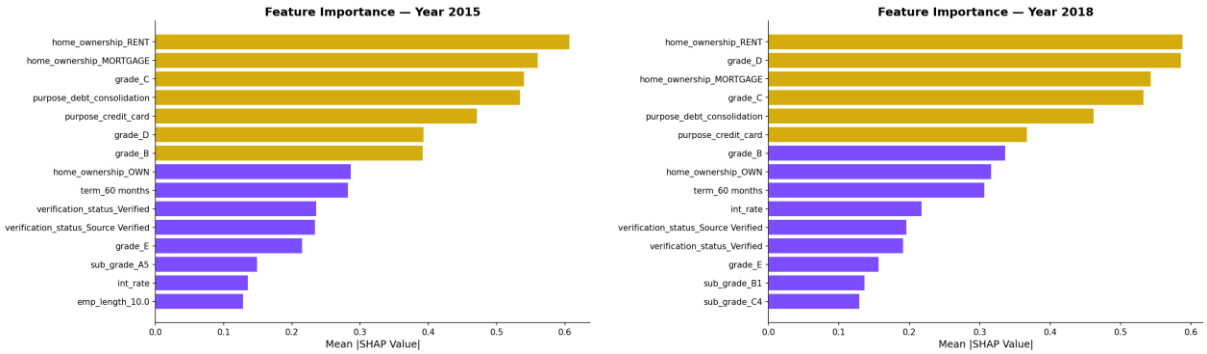


Figure 5.7: Top SHAP features of the static model on 2015 (left) and 2018 (right).

Figure 5.7 shows the top features directly. Home ownership (renting or holding a mortgage), loan grade and the debt consolidation purpose dominate in both years, which matches domain intuition. The change between the two panels is modest but meaningful: in 2018 grade D moves to second place and interest rate is ranked ninth. For RQ4, the conclusion is that explanations drift far less than accuracy, which is reassuring for reviewers, and that changes in the top-ten membership agree with the feature-level drift signal, so explanation monitoring can serve as an interpretable second opinion alongside the detectors.

5.8 Implementation issues identified during testing

Several defects were found while auditing the outputs, and each changed a reported result. They are listed here because correcting them is part of the contribution of the work.

- **Per-window encoding.** Encoding each yearly window separately changed the dropped baseline category between years, so identically named columns meant different things. Fixed by fitting one category schema on the full data and sharing it across windows.
- **Unfair evaluation.** The original retraining routine scored the adaptive model on a mixed split of several years and the static model on the full target year. Replaced by the shared held-out test split described in Section 4.6.
- **Constant Jaccard index.** Both SHAP vectors were reindexed onto a shared feature order before the top ten were taken, which forced the two top-ten sets to be identical and fixed the Jaccard

index at 1.0. Fixed by sorting each vector independently; the real values are 0.82 in 2017 and 2018.

- **Threshold artefact.** F1 at 0.5 favoured the static model. Fixed by adding PR-AUC and a validation-tuned threshold, which reversed the comparison.
- **Serialisation and scale.** NumPy boolean results from the KS test could not be written to JSON, and exact SHAP on the full reference window was intractable. Fixed by casting to native types and by sampling 3,000 rows per window with a tree-path-dependent fallback.
- **Dependency conflict.** An imbalanced-learn release was incompatible with the installed scikit-learn version and broke SMOTE. Resolved by pinning compatible versions in the project environment.

5.9 Interactive dashboard

The Streamlit dashboard makes every result above available to a non-programmer. Beyond the static charts, its live drift check page lets a user choose a built-in year or upload a new file of loans. It then runs PSI, KS, JS and the classifier two-sample test against the 2015 reference in roughly ten to forty seconds, scores the static and adaptive models, and reports the live AUC and PR-AUC lift. For the built-in years it uses the same held-out split as the experiments, so the 2018 result it shows (0.6691 to 0.7222) matches Table 5.5 exactly.

6 Summary and Future plan of work

6.1 Summary of the work

This project built and evaluated an adaptive credit risk scoring pipeline on four years of real lending data. MTP-1 established the data pipeline, the baseline models and univariate drift monitoring, and measured how fast a static model decays. MTP-2 added multivariate drift detection, a leakage-free retraining protocol, a comparison of retraining strategies, threshold-aware evaluation, explanation monitoring and a dashboard. The main findings are:

- A credit model trained once loses 9.8% of its AUC-ROC over three years, most of it in the first year.
- The drift is largely multivariate. A classifier two-sample test separates 2016 loans from 2015 loans with an AUC of 0.977, while PSI flags no single feature that year; a PSI-only trigger would have missed the retrain.
- Drift-triggered retraining, judged on identical held-out loans, improves AUC-ROC by 0.018 to 0.053 and PR-AUC by 0.040 to 0.074, and the benefit grows with the amount of drift.
- Recency beats volume: the most-recent-year model is best and the expanding window is worst among the retraining strategies.
- Global explanations remain stable, while changes in the top-ten features, led by interest rate, follow the feature-level drift.

6.2 Limitations

The conclusions should be read with the following limits in mind. The public data ends in 2018, so the far sharper drift of the COVID-19 period could not be tested. Only one market and one lender are studied. Yearly windows are coarse compared with the monthly monitoring a lender would run, and they provide only three drift events. Because labels require loans to mature, recent years contain fewer and possibly unrepresentative loans. SMOTE distorts predicted probabilities, and no calibration step was added, so the scores should not be read directly as default probabilities. Finally, because drift fired every year, the study cannot show that the trigger saves retrains over a fixed yearly schedule.

6.3 Future plan of work

- Move from yearly to monthly or quarterly windows and add streaming detectors such as ADWIN on the feature distributions, so that drift is caught earlier and the trigger can be shown to skip unnecessary retrains.
- Treat the window length as a parameter chosen on validation data, and test weighted schemes that keep older loans with a lower weight instead of discarding them.
- Add probability calibration (isotonic or Platt scaling) after resampling so that scores can be used for pricing and expected-loss estimates.
- Extend the evaluation to other credit datasets and shock periods, and add fairness monitoring across borrower groups following recent work on fair credit scoring under drift [18].

Appendix (if any)

A.1 Reproducing the results

All code, configuration and generated reports are kept in the project repository. The pipeline is run in the following order from the project root, using the project's Python virtual environment:

- Preprocessing: src/data/preprocess.py builds the cleaned table and the four yearly windows.
- Phase 2: run_phase2.py trains the three baseline models on 2015 and logs them to MLflow.
- Phase 3: run_phase3.py runs PSI, KS, JS and C2ST and writes the drift report.
- Phase 4: run_phase4.py trains the sliding, naive-recent and expanding models, tunes thresholds and computes SHAP stability.
- Phase 5: run_phase5.py runs the backtest, draws the figures and regenerates the text summary.
- Dashboard: streamlit run dashboard/app.py opens the results and the live drift check at http://localhost:8501.

A.2 Configuration and environment

The main settings are held in configs/config.yaml: the feature lists, the 80/20 split with seed 42, the PSI warning and critical thresholds of 0.1 and 0.2, the KS significance level of 0.05, a sliding window of three years and a SHAP sample of 3,000 rows. The environment uses scikit-learn 1.5.2, XGBoost 2.0.3, LightGBM 4.3.0, imbalanced-learn 0.13.0, NumPy 1.26.4, MLflow 2.12.1 and Streamlit 1.39. A full run of Phase 4, which fits nine resampled XGBoost models and computes SHAP values, takes about seventeen minutes on a standard laptop.

A.3 Output artefacts

The reports directory contains backtest_results.csv, final_comparison.csv, strategy_comparison.csv, drift_methods_comparison.csv, shap_stability_report.csv, static_degradation.csv, drift_report.json and final_summary.txt. The plots directory contains every figure used in Chapter 5, and the models directory holds the saved static and adaptive models together with the tuned decision thresholds.

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